QUANTITATIVE METHODS 1

Worksheet 6 – Index numbers

Exercises

Exercise 1 – Elementary indices, rebasing and properties

A service costs 80 euros in 2023, 92 euros in 2024 and 101.20 euros in 2025. Calculate the three base-100 indices, verify circularity, calculate the reverse index and verify reversibility, rebase the series to 2024=100, and interpret index 126.5 correctly.

Exercise 2 – Laspeyres, Paasche, Fisher and factor reversal

	p_0	q_0	p_1	q_1
Product X	4	100	5	90
Product Y	10	40	12	50

Calculate the four aggregates, Laspeyres, Paasche and Fisher price indices, the corresponding quantity indices, the value index, and verify Fisher factor reversal on a base-100 scale.

Exercise 3 – Chain indices and deflation

Annual link indices are 105, 108 and 97. Calculate the cumulative chain index and interpret it. Explain why annual changes must be compounded rather than added. Then deflate nominal revenue of 2.4 million euros when the price index equals 120.